Clipcademy review study · Form A

Participant code: _____ Date: _____

10 questions. Circle one letter for each. Don't use notes or your phone. Your answers don't affect your grade.

1. A template strand runs 3′ → 5′ from left to right. In which direction does the new strand grow along it?

(A) Right to left
(B) Left to right
(C) Both ways from the middle
(D) It can't grow along this strand

2. A new DNA strand reads 5′-AGT-3′. The next nucleotide added is C. What does the strand read now?

(A) 5′-CAGT-3′
(B) 5′-ACGT-3′
(C) 5′-AGCT-3′
(D) 5′-AGTC-3′

3. A tube holds a single-stranded DNA template, DNA nucleotides and DNA polymerase, but no primase and no primer. What happens?

(A) DNA is made in the 3′ to 5′ direction
(B) DNA polymerase makes its own primer first
(C) No new DNA is made
(D) DNA ligase starts the new strand

4. A wrong base slipped past proofreading, and the DNA has now been fully copied. What can still correct it?

(A) Mismatch repair proteins
(B) DNA polymerase proofreading
(C) Primase replacing the base
(D) Nothing, the error is permanent

5. UV light has bonded two neighbouring T's on one DNA strand. Why is this a problem for the cell?

(A) The two strands fall apart
(B) A base is lost from the strand
(C) A bulge blocks normal copying
(D) The T's turn into C's

6. In UV-damage repair, what happens right after the damage site is recognized?

(A) New bases are put in the gap
(B) The backbone is sealed shut again
(C) An RNA primer is added to it
(D) The damaged piece is cut out

7. An NTP carrying G enters RNA polymerase's active site. The next base on the template strand is C. What happens?

(A) It is joined to the growing RNA
(B) It isn't joined, because it doesn't match
(C) It pairs with the coding strand instead
(D) It waits for the coding strand to be read

8. A mutation destroys a gene's promoter. What is the most direct effect?

(A) The ribosome can't find the start codon
(B) RNA polymerase can't start on the gene
(C) DNA polymerase can't copy the gene
(D) The tRNAs can't carry amino acids

9. An mRNA codon reads CGA. Which tRNA anticodon pairs with it? (Bases are listed in the order they pair, like AUG and UAC in the notes.)

(A) CGA
(B) AGC
(C) GCA
(D) GCU

10. A tRNA holding the growing protein sits in the P site, and a matching charged tRNA has just bound in the A site. What happens next?

(A) The A-site tRNA leaves from the E site
(B) The P-site tRNA picks up an amino acid
(C) A peptide bond joins their amino acids
(D) A new tRNA enters the P site

Moderator only: score _____ / 10